$$= 4\left[\int_0^1\int_0^1 x^2y\,dx\,dy + \int_0^1\int_0^1 xy^2\,dx\,dy\right]$$

$$= 4\left[\int_0^1\left(\int_0^1 x^2\,dx\right)ydy + \int_0^1\left(\int_0^1 y^2\,dy\right)xdx\right]$$

$$= 4[\int_0^1 \frac{1}{3}y\,dy + \int_0^1 \frac{1}{3}x\,dx\,]$$

$$= \frac{4}{3}\left[\int_0^1 y\,dy + \int_0^1 x\,dx\right]$$

$$= \frac{4}{3}[\frac{1}{2}+\frac{1}{2}] = \frac{4}{3} \qquad \ldots (1)$$

$$E(X) = \int_{-\infty}^{\infty} xf(x)\,dx$$

$$= \int_0^1 x \times 2x\,dx$$

$$=2\int_0^1 x^2\,dx$$

$$E(X) = 2\left[\frac{1}{3}\right] = \frac{2}{3}$$

$$\mathrm{E(Y)} = \int_{-\infty}^{\infty} yf(y)\,dy$$

$$= \int_0^1 y \times 2y\,dy$$

$$=2\int_0^1 y^2\,dy$$

$$E(Y) = 2\left[\frac{1}{3}\right] = \frac{2}{3}$$

$$\mathrm{E(X)}+E(Y) = \frac{2}{3}+\frac{2}{3} = \frac{4}{3} \qquad \ldots (2)$$

From 1&2

$$E(X+Y) = E(X)+E(Y)$$

9.7.2 Multiplication Theorem on Expectation

Discrete random variable:

Statement: If X and Y are two independent variables then $E(XY) = E(X)\,E(Y)$